Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Full-Stack Developer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and mobile applications for client businesses via Upwork and Fractal, specializing in full-stack architecture, e-commerce infrastructure, and automated testing pipelines.
- **Order Procurement & Management Solution** | *NextJS, NodeJS, PostgreSQL, TypeScript*
  - Architected a full-stack enterprise order portal utilizing NextJS, ExpressJS, TypeScript, and PostgreSQL with session-based authentication and RESTful APIs.
  - Designed a dual-tier interface comprising a customer ordering portal with real-time inventory listings and an office administrative suite supporting employee task dispatch, audit logging, and recurring order schedules.
  - Developed an automated order ingestion pipeline parsing incoming WhatsApp messages into structured database records, eliminating manual entry and accelerating order processing throughput.
  - Designed relational PostgreSQL schemas with optimized indexing and relational integrity constraints to handle high-volume transactions efficiently.

Software Developer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Flutter, Django, OpenCV, C* **(Live link)**
  - Architected a native C/C++ bridge using `dart:ffi` to port OpenCV Scale-Invariant Feature Transform (SIFT) keypoint algorithms into Flutter, enabling single-codebase cross-platform computer vision across Android and iOS.
  - Engineered a real-time camera feed pipeline that normalized OS frame buffers and aspect ratios, converting raw video streams into C structs with zero memory leaks and 30+ FPS performance.
  - Implemented real-time keypoint extraction and object recognition, transmitting match coordinates asynchronously to Dart for dynamic augmented reality UI overlay rendering.
  - Developed responsive mobile interfaces in Flutter from Figma designs and built RESTful backend APIs in Django to power location-aware spatial discovery features for alpha release.
- **Kia National Conference App** | *Flutter*
  - Engineered an event management mobile app in Flutter for 500+ attendees at the Kia National Dealer Conference in Bangkok, featuring real-time registration, attendee directories, and interactive venue navigation.

Technical Skills

Languages & Systems: Python, JavaScript/TypeScript, C++, Dart, C, SQL, R, Mathematica
Frameworks & Databases: NodeJS, ExpressJS, NestJS, Django, PostgreSQL, Flutter, React, NextJS, React Native
Computer Vision & Data Science: OpenCV (SIFT), NumPy, Pandas, Scikit-learn, Seaborn, Matplotlib
Developer Tools & Cloud: AWS, Git, Linux, Bash, Docker, Vim, LaTeX, Agentic Development, MCPs, LLM Pipeline & Context Engineering, n8n

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)